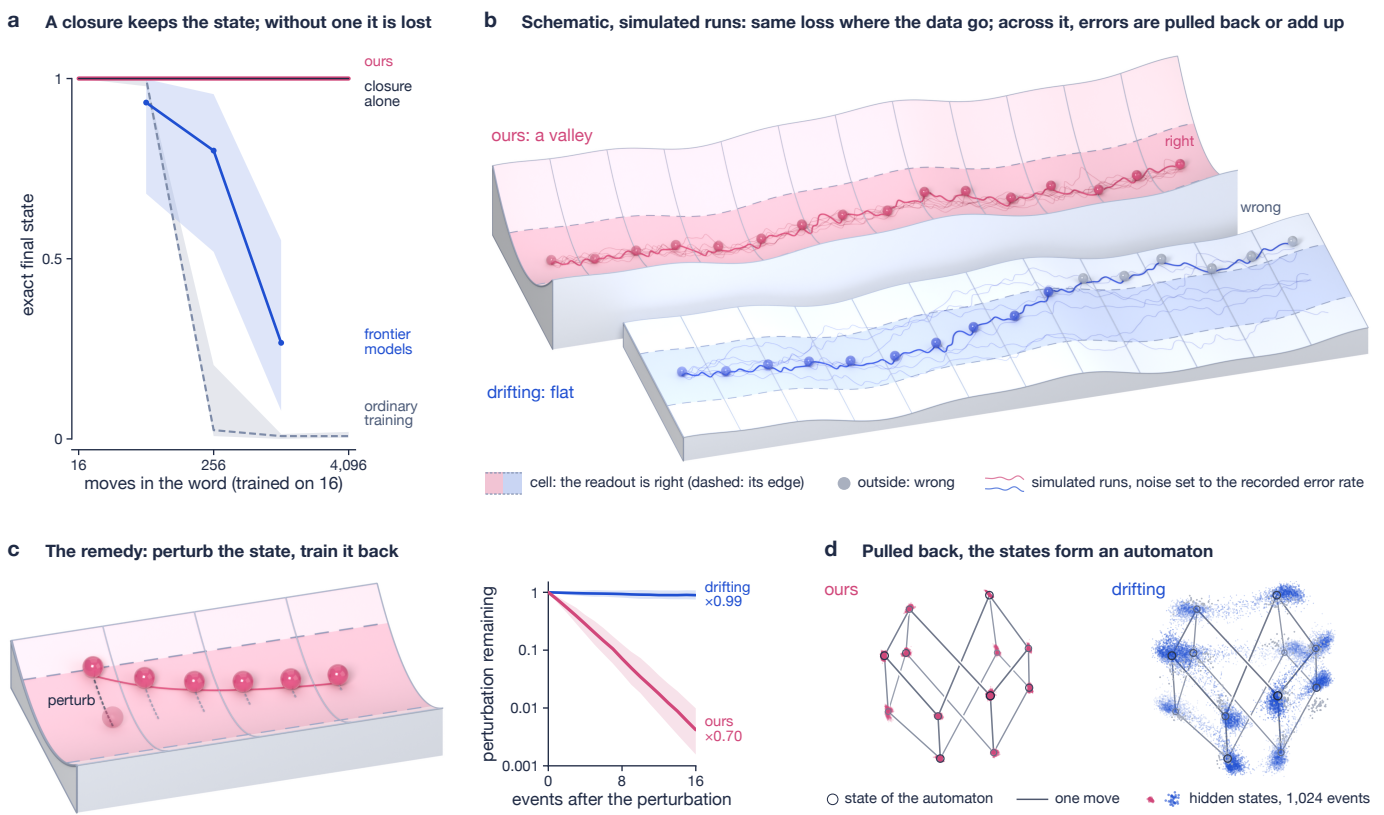

a
A closure keeps the state; without one it is lost
ours
closure alone
frontier models
ordinary training
exact final state
1
0.5
0
16
256
4,096
moves in the word (trained on 16)
b
Schematic, simulated runs: same loss where the data go; across it, errors are pulled back or add up
ours: a valley
right
wrong
drifting: flat
cell: the readout is right (dashed: its edge)
outside: wrong
simulated runs, noise set to the recorded error rate
c
The remedy: perturb the state, train it back
perturb
perturbation remaining
1
0.1
0.01
0.001
drifting ×0.99
ours ×0.70
0
8
16
events after the perturbation
d
Pulled back, the states form an automaton
ours
drifting
state of the automaton
one move
hidden states, 1,024 events